

Applying Exponent LawsQuestion Bank 3.05 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 4, rewrite each expression as an equivalent expression with a positive exponent.**

1. $\left(\frac{3}{4}\right)^{-3}$

$\left(\frac{4}{3}\right)^3$

2. $\left(-\frac{7}{8}\right)^{-2}$

$\left(-\frac{8}{7}\right)^2$

The sign stays with the fraction.

3. $\left(-\frac{4}{5}\right)^{-4}$

$\left(\frac{5}{-4}\right)^4$

4. $\left(\frac{22}{7}\right)^{-7}$

$\left(\frac{7}{22}\right)^7$

Matching.

5. Match each expression on the left with the equivalent expression on the right by writing the letter.

a. $\left(-\frac{11}{10}\right)^{-3}$

A. $\left(-\frac{11}{10}\right)^3$

B. $\left(\frac{10}{11}\right)^3$

C. $\left(\frac{11}{10}\right)^3$

D. $\left(-\frac{10}{11}\right)^3$

D

b. $\left(\frac{11}{10}\right)^{-3}$

B

c. $\left(-\frac{10}{11}\right)^{-3}$

A

d. $\left(\frac{10}{11}\right)^{-3}$

C

For questions 6 through 9, generate an equivalent expression with one base and one positive exponent. Show the laws you use.

6. $\left(-\frac{14}{17}\right)^{-2} \cdot \left(-\frac{14}{17}\right)^{-4}$

$$\left(-\frac{14}{17}\right)^{-6} = \left(-\frac{17}{14}\right)^6$$

Product of powers, then flip.

7. $\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$

$$\left(\frac{5}{8}\right)^2$$

$-2 - (-4) = 2$; already positive.

8. $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$

$$\left(\frac{2}{3}\right)^9$$

$(5 - 2) \cdot 3$.

9. $\left(\frac{2.5^5}{2.5^3}\right)^{-1}$

$$2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2$$

Rewrite the decimal as a fraction before flipping.

Equivalent or not.

10. Consider the expression $\frac{\left(\frac{1}{6}\right)^3}{6^{-5}} \cdot 6^4$. For each expression below, decide whether it is equivalent, and justify with a law of exponents.

a. $\frac{6^{-3}}{6^{-5}} \cdot 6^4$

Yes — $\left(\frac{1}{6}\right)^3 = 6^{-3}$, so this is the same expression; value 6^6 .

b. $6^{-8} \cdot 6^4$

No — the quotient gives 6^2 , not 6^{-8} ; this is 6^{-4} .

c. $\left(\frac{1}{6}\right)^{-2} \cdot 6^4$

Yes — $\left(\frac{1}{6}\right)^{-2} = 6^2$, and $6^2 \cdot 6^4 = 6^6$.

d. 6^6

Yes — it is the value of the expression written as one power.

11. Which is the value of $\left(-\frac{2}{3}\right)^{-3}$?

☒ A. $-\frac{27}{8}$

☐ B. $\frac{27}{8}$

☐ C. $-\frac{8}{27}$

☐ D. $\frac{8}{27}$

A

$$(-3/2)^3 = -27/8.$$

12. Find the value of n that makes each statement true.

a. $\left(\frac{3}{5}\right)^n = \frac{25}{9}$

$n = -2$

$$(5/3)^2 = 25/9.$$

b. $\left(\frac{1}{2}\right)^n = 8$

$n = -3$